

Multi-label Scandinavian Language Identification (SLIDE)

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RESOURCEFUL-2025



* Equal contribution.

- 1 Introduction
- 2 Data
- 3 Evaluation
- 4 Our approach: BERTs (SLIDE-xs, SLIDE-s, SLIDE-base)
- 5 Results
- 6 Discussion
- 7 Appendix

Why multi-label Scandinavian language identification?

Identifying closely related languages at sentence level is difficult:

<i>En dag i livet</i>	→	DA	NB	NN	SV
<i>Jag vill ha dig</i>	→	DA	NB	NN	SV
<i>Jeg er hvalrossen</i>	→	DA	NB	NN	SV
<i>Denne fuglen har flydd</i>	→	DA	NB	NN	SV

Sentences valid in multiple Scandinavian languages are common: 5% of the test dataset and 16% of the sentences shorter than 6 words

Introduction

Why should we care about sentence-level LID?

- ▶ As modern language models are mostly pretrained on web crawls ([Liu et al., 2019](#)), ([Touvron et al., 2023](#)), texts of any length may occur in the pretraining data
- ▶ Code switching: a single text may contain sentences in different languages

Introduction

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Main contributions

- ▶ A multi-label evaluation dataset
- ▶ A suite of LID models
- ▶ A novel method of silver-labeling a LID dataset

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Why just another LID dataset?

most existing LID corpora rely on the source of a text: if a sentence is retrieved from a Danish newspaper, it is assumed to be only Danish. This approach doesn't work for similar languages ([Goutte et al., 2016](#); [Keleg and Magdy, 2023](#))

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Initial data sources

the Universal Dependencies 2.14 treebanks ([Nivre et al., 2016, 2020](#)) with their train/dev/test splits

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Data

Manual inspection of dev and test splits

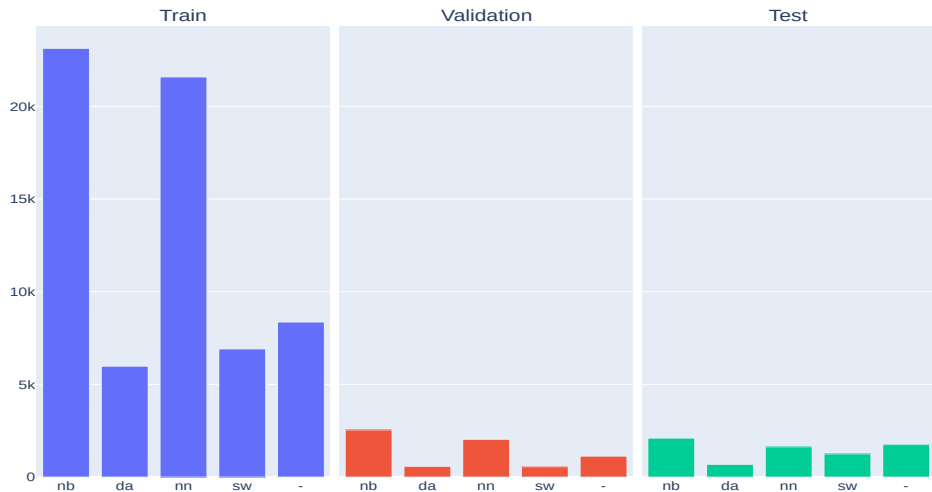
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Machine translation silver-labeling of the train split

'En dag i livet'

- ▶ Bokmål: En dag i livet
- ▶ Nynorsk: Ein dag i livet
- ▶ Danish: En dag i livet
- ▶ Swedish: En dag i livet

NorMistral-11b ([Samuel et al., 2024](#)), further fine-tuned on Tatoeba evaluation set ([Tiedemann, 2020](#)) in all translation directions between Bokmål, Danish, Nynorsk and Swedish



Number of sentences per language (kept as in the original treebanks)

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Loose accuracy

- ▶ a prediction is considered correct if intersection between predictions and gold labels is not empty
a model that always predicts all four languages would get 100%

Strict accuracy

- ▶ exact match between the predicted and gold labels sets
a model that always predicts all four languages would get as many % as much share all-four instances is in the data

Per-language F1-scores

- ▶ a true positive prediction happens if and only if the respective language is present both in the set of predicted labels and in the set of gold labels

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Our approach: BERTs (SLIDE-xs, SLIDE-s, SLIDE-base)



Base model selection

Model	Loose accuracy	Exact-match accuracy	Macro F_1
XLM-RoBERTa-base (Conneau et al., 2020)	96.8	94.6	95.4
DistilBERT-base (Sanh et al., 2019)	96.5	94.5	95.2
ScandiBERT(Snæbjarnarson et al., 2023)	97.6	95.9	96.6
NorBERT3-base (Samuel et al., 2023)	98.6	96.4	97.0

Base model selection We made our choice based on the validation data split, the metrics in this table, given in percent, are for the test split. F_1 is per-language exact match. NorBERT3 refers to the same model as SLIDE.

Our approach: BERTs (SLIDE-xs, SLIDE-s, SLIDE-base)



Data augmentation and normalization

Basic idea: ensure that '*Oslo är Norges huvudstad*' will not be labeled as Norwegian

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- ▶ Regular expression normalization - normalize URLs, email addresses, and numbers into the following special symbols: $\langle \text{URL} \rangle$, $\langle \text{mail} \rangle$ and $\langle \text{num} \rangle$

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- ▶ Alphabet variations - adding Swedish sentences containing the Danish–Norwegian letters and Danish and Norwegian sentences containing the Swedish letters (e.g., in proper names and in the context of quotations)

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- ▶ Named entity swaps - (NER) on the training data using spaCy; randomly swap the recognized entities with other entities from the same category

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Data augmentation

Alterations	Loose accuracy	Exact-match accuracy
Augmentation + Regex normalization	98.6	96.4
Augmentation	98.4	96.3
Regex normalization	98.4	96.2
NER	98.7	95.5
Base	98.3	96.2

Ablation study Impact of data augmentation and regular expression normalization on SLIDE-base measured by test set performance. "Augmentation" refers to punctuation and alphabet augmentation, "Regex" refers to regular expression normalization, "NER" refers to named entity swaps and "Base" is neither of the above.

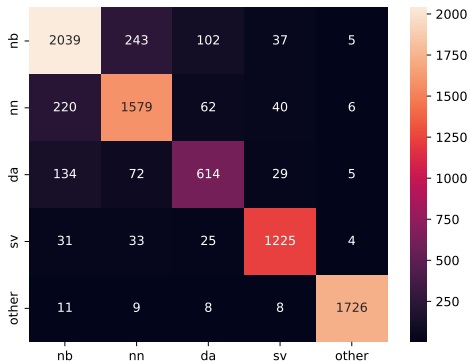
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Results

Model	Loose accuracy	Exact-match accuracy	NB F ₁	DA F ₁	NN F ₁	SV F ₁	Other F ₁	Runtime ms/sample
BASELINES								
gpt2-lang-ident	61.2	58.9	47.0	24.0	36.9	83.6	86.2	52.07
FastText-176*	80.5	77.7	72.6	66.0	55.7	92.7	93.5	0.01
NLLB-218*	95.3	91.6	93.0	85.9	89.0	96.8	93.6	0.08
NB-Nordic-LID*	83.3	80.6	85.0	67.0	84.8	89.7	70.2	0.02
OpenLID*	94.2	90.2	91.5	82.6	88.7	95.7	93.3	0.08
GlottLID*	97.2	93.4	93.5	89.5	89.4	97.9	98.1	0.51
<i>Heliport (HeLI-OTS)</i>	96.5	92.6	90.9	89.0	91.2	97.6	97.2	0.02
OUR MODELS								
SLIDE-x-small (15M)	97.8	95.7	97.5	90.4	96.2	98.0	98.7	13.22
SLIDE-small (40M)	98.1	95.7	97.7	89.9	96.3	98.0	99.1	19.70
SLIDE-base (123M)	98.6	96.4	98.1	92.0	97.1	98.6	99.4	38.41

* shows which baselines use FastText. Heliport is the only multilabel one

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SLIDE-base confusion matrix

Sources of Norwegian errors: spelling variations, ambiguity

- ▶ 'høyre' - Nynorsk ('hear'), Bokmål ('right');
- ▶ 'I alle år har vi fått høyre at med dagens forbruk er det olje nok for mange ti år.' (Nynorsk) misclassified as Bokmål
- ▶ 'I den nye designen er høgre og venstre spalte på framsida til nettavisa fjerna.' (Bokmål, Nynorsk) misclassified as Nynorsk only

Conclusion

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- ▶ using machine translation for creating a silver multi-label training dataset from a single-label one has proved to be efficient
- ▶ no clear answer how much data preprocessing/data augmentation makes the model most robust
- ▶ future work and the right way to solve the task: multilabel with machine translation
GlotLID's dataset (3.9M samples for Norwegian Bokmål only); tokenize in a Scandinavian-friendly way; train FastText embeddings; train a multilabel classifier on top of it. An open question is how many 'other' data is needed



We would like to thank Helene Bøsei Olsen and Karoline Sætrum for their work on annotating the initial version of the test set.

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GlottLID is a strongest baseline and faster than a BERT: but...

- ▶ >2000 languages is a bottleneck for the runtime
- ▶ single-labeling
- ▶ solution: train an own MLP on top of GlottLID embeddings
- ▶ still a problem: OOV words; data normalization/augmentation did not help (probably also because of OOV)

Results

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BASELINES								
GlottLID*	97.2	93.4	93.5	89.5	89.4	97.9	98.1	0.51
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OUR MODELS								
SLIDE-fast	95.7	93.4	94.5	90.2	92.4	97.5	96.4	0.16
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SLIDE-small	98.1	95.7	97.7	89.9	96.3	98.0	99.1	19.70
SLIDE-base	98.6	96.4	98.1	92.0	97.1	98.6	99.4	38.41

Did we overfit to UD?

Model	3K test split	15K test split
SLIDE-base	92.7	95.3
SLIDE-fast	85.4	88.5
GlottLID	93.0	95.7

As ([Haas and Derczynski, 2021](#)) is a single-label dataset, we consider a prediction to be correct, if one of the predicted languages is correct.

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- ▶ 'ou di be t aatm ne enwadi' - Swedish, 'atahualpa yupanqui' - Danish, 'tromssan ruijan-suomalainen yhdistys' - Nynorsk

Confusion with 'other'

- ▶ proper names ('kruvi: *Karl Marx*') (50% of 'other' instances misclassified as Scandinavian)
- ▶ English (30% of 'other' instances misclassified as Scandinavian)
- ▶ loanwords: server med pastasalat med bakte grønsaker og tsatsiki til
- ▶ 'Va shiaulteyr er ny skeabey harrish boayrd.' (Manx)

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